

Lista 4 de Calculo II Jeann Victor

1) Calcule as integrais das potências de seno e cosseno

a) $\int \sin x \cos^2 x \, dx$ função é ímpar (por isso) ...
 $\int \sin x \cdot (\cos x)^2 \, dx$ pela regra da substituição

$$u = \cos x, du = -\sin x \, dx$$

$$\int (\cos)^2 \cdot \sin x \, dx \rightarrow \int u^2 \cdot (-du) \rightarrow -u^3/3 + C$$

$$\frac{\cos^3 x}{3} + C \text{ é a primitiva.}$$

b) $\int_{-\pi/2}^{\pi/2} \cos^2 x \, dx$

$$\int \frac{1 + \cos(2x)}{2} \, dx \rightarrow \int \frac{1}{2} \, dx + \int \frac{\cos(2x)}{2} \, dx$$

$$\frac{1}{2} \int dx + \frac{1}{2} \int \cos(2x) \, dx \rightarrow \frac{1}{2} \cdot x + \frac{1}{2} \cdot \frac{\sin(2x)}{2} \rightarrow \left(\frac{x}{2} + \frac{\sin(2x)}{4} \right)$$

A primitiva é: $x/2 + \sin(2x)/4$

Intervalo de $-\pi/2$ a $\pi/2$

$$\left(\frac{\pi}{2} + \frac{\sin(\pi)}{4} \right) - \left(-\frac{\pi}{2} + \frac{\sin(-\pi)}{4} \right)$$

$$\frac{\pi}{2} + \frac{\pi}{2} \rightarrow \frac{2\pi}{2} \rightarrow \pi$$

c) $\int_{-\pi/2}^{\pi/2} \sin^3 x \cdot \cos^2 x \, dx$

A função é ímpar, portanto = 0.

d) $\int_0^{2\pi} \cos^3 x \, dx \rightarrow \int \cos x \cdot \cos^2 x \rightarrow \int \cos x \cdot (1 - \sin^2 x)$

$$\int \cos x - \cos x \cdot \sin^2 x \rightarrow \int \cos x - \int \cos x \cdot \sin^2 x$$

$$\int \cos x \cdot \sin^2 x \, dx = u = \sin x, du = \cos x \, dx$$

$$\int u^2 \cdot \cos x \, dx \rightarrow \int u^2 \cdot du \rightarrow u^3/3 \rightarrow \sin^3 x/3$$

$$\sin x - \sin^3 x + C$$

Intervalo de 0 a 2π

$$\frac{\sin^3 2\pi - \sin^3 0}{3} - \left(\frac{\sin^3 0 - \sin^3 0}{3} \right) \rightarrow 0$$

$$e) \int_0^{\pi/2} \sin^4 x \, dx \rightarrow (\sin^2 x)^2 \, dx \rightarrow \left(\frac{1 - \cos(2x)}{2} \right)^2$$

$$\frac{1}{4} (1 - 2\cos(2x) + \cos^2(2x))$$

$$\frac{1}{4} \int (1 - 2\cos(2x) + (1 + \cos(4x))) \, dx$$

$$\left(\frac{1}{4} - \frac{\cos(2x)}{2} + \frac{1}{8} + \frac{\cos(4x)}{8} \right) dx \rightarrow \frac{3}{8} - \frac{1}{2} \cos(2x) + \frac{1}{8} \cos(4x)$$

$$\int \frac{3}{8} \, dx - \frac{1}{2} \int \cos(2x) \, dx + \frac{1}{8} \int \cos(4x) \, dx$$

$$\int \cos(2x) \, dx = u = 2x, \, du = 2 \, dx \rightarrow du/2 = dx$$

$$\int \cos(u) \cdot \frac{1}{2} \, du \rightarrow \frac{1}{2} \sin(u) \rightarrow \frac{1}{2} \sin(2x) + C$$

$$\dots \int \cos(4x) \rightarrow \frac{1}{4} \sin(4x)$$

$$\frac{3}{8} x - \frac{1}{4} \sin(2x) - \frac{1}{32} \sin(4x) + C$$

Intervalo de $(0, \pi/2)$

$$\left(\frac{3}{8} \cdot \frac{\pi}{2} - \frac{1}{4} \sin \pi - \frac{1}{32} \sin(2\pi) \right) - \left(\frac{3}{8} \cdot 0 - \frac{1}{4} \sin 0 - \frac{1}{32} \sin(0) \right)$$

$$\frac{3\pi}{16}$$

$$f) \int \sin^2 x \cdot \cos^3 x \, dx$$

$$16. \int \sin^2 x \cdot \cos^3 x \, dx \rightarrow \sin x \cdot \cos x = (\sin(2x))/2 \rightarrow$$

$$16. \int \frac{\sin^2(2x)/2}{2} dx \rightarrow \frac{1}{4} \cdot \int 1 - \cos(4x) \rightarrow 2 \cdot \int (1 - \cos(4x)) dx$$

$$2 \cdot \int 1 \, dx - 2 \cdot \int \cos(4x) \, dx$$

$$\int \cos(4x) \, dx \rightarrow u = 4x, du = 4 \, dx, du/4 = dx$$

$$\int \cos(u) \, du \cdot \frac{1}{4} \rightarrow \frac{1}{4} \sin(4x)$$

$$2x + 1/2 \sin(4x) \rightarrow 2x + \frac{\sin(4x)}{2} + C$$

$$g) \int \sin^5 x \cdot \cos^3 x \, dx$$

$$\int (\sin(x) \cdot (\sin^4 x) \cdot \cos^3 x) \, dx$$

$$\int (\sin(x) \cdot (1 - \cos^2 x)^2 \cdot \cos^3 x) \, dx$$

$$\int (\cos^3 x - 2 \cdot \cos^5 x + \cos^7 x) \cdot \sin(x) \, dx$$

$$u = \cos x, du = -\sin(x) \, dx \rightarrow -du = \sin(x) \, dx$$

$$\int (u^3 - 2 \cdot u^5 + u^7) \cdot -du$$

$$-\int u^3 \, du + 2 \cdot \int u^5 \, du - \int u^7 \, du$$

$$-\frac{u^4}{4} + \frac{2u^6}{6} - \frac{u^8}{8} \rightarrow -\frac{\cos^4 x}{4} + \frac{\cos^6 x}{3} - \frac{\cos^8 x}{8} + C$$

$$h) \int \sin^3 x \cdot \sqrt{\cos x} \, dx$$

$$\int (\sin^2 x \cdot \sin x \cdot \cos^{1/2} x) \, dx$$

$$\int (1 - \cos^2 x) \cdot \sin x \cdot \cos^{1/2} x \, dx$$

$$\int (1 - \cos^2 x) \cdot \cos^{1/2} x \cdot \sin x \, dx \rightarrow \int (\cos^{1/2} x - \cos^{5/2} x) \cdot \sin(x) \, dx$$

$$u = \cos x, du = -\sin x \, dx, -du = \sin x \, dx$$

$$\int (u^{1/2} - u^{5/2}) \cdot -du \rightarrow \int u^{1/2} \, du - \int u^{5/2} \, du$$

$$\frac{u^{3/2}}{3/2} - \frac{u^{7/2}}{7/2} \rightarrow -\frac{2u^{3/2}}{3} + \frac{2u^{7/2}}{7} + C$$

2) Calcule as integrais das potências de secante e tangente abaixo

a) $\int \sec^2 x \cdot \tan x \, dx$

$\int (\sec x \cdot \sec x \cdot \tan x) \, dx$, $u = \sec x$, $du = \sec x \tan x \, dx$

$\int u \cdot du \rightarrow \frac{u^2}{2} + C$

b) $\int \sec^3 x \cdot \tan x \, dx$

$\int (\sec^2 x \cdot \sec x \cdot \tan x) \, dx$, $u = \sec x$, $du = \sec x \tan x \, dx$

$\int u^2 \cdot du \rightarrow \frac{u^3}{3} + C$

c) $\int \sec^4 x \, dx$

$\int (\sec^2 x \cdot \sec^2 x) \, dx \rightarrow \int (1 + \tan^2 x) \cdot \sec^2 x \, dx$

$\int (\sec^2 x + \sec^2 x \cdot \tan^2 x) \, dx$

$\int \sec^2 x \, dx + \int \sec^2 x \cdot \tan^2 x \, dx$

$\int \sec^2 x \, dx \rightarrow \tan x$

$\int \sec^2 x \cdot \tan^2 x \rightarrow u = \tan x$, $du = \sec^2 x \, dx$

$\int u^2 \cdot du \rightarrow \frac{u^3}{3} \rightarrow \frac{\tan^3}{3} + C$

$\tan x + \frac{\tan^3}{3} + C$

d) $\int \tan^4 x \, dx$

$\int (\tan^2 x \cdot \tan^2 x) \, dx \rightarrow \int (\sec^2 x - 1) \cdot \tan^2 x \, dx$

$\int \sec^2 x \cdot \tan^2 x - \tan^2 x \, dx \rightarrow \int \sec^2 x \cdot \tan^2 x \, dx - \int \tan^2 x \, dx$

1º $\int \sec^2 x \cdot \tan^2 x \, dx = u = \tan x$, $du = \sec^2 x \, dx$

$\int u^2 \cdot du \rightarrow \frac{u^3}{3} \rightarrow \frac{\tan^3}{3}$

2º $\int \tan^2 x \, dx \rightarrow \int (\sec^2 x - 1) \, dx \rightarrow \int \sec^2 x \, dx - \int dx$

$\tan x - x$

Juntando.

$\frac{\tan^3}{3} - \tan x + x + C$

③ Um pêndulo com comprimento L forma um ângulo máximo de θ_0 com a vertical. Usando a 2ª Lei de Newton, pode ser mostrado que o período T é dado por:

$$T = 2\sqrt{\frac{L}{g}} \int_0^{\pi/2} dx / \sqrt{1 - a^2 \sin^2 x}$$

onde $a = \sin(\theta_0/2)$ e g é a aceleração da gravidade.

Considerando a aproximação $(1 - a^2 \sin^2 x)^{-1/2} \approx 1 + \frac{a^2 \sin^2 x}{2}$, mostre que $\omega \approx L \cdot \omega_0 (1 + \frac{a^2}{4})$, onde $\omega = 2\pi/T$ é a frequência angular do movimento e $a^{2/4}$ é o primeiro termo de correção de g para oscilações ligeiramente maiores do que as consideradas em um pêndulo simples.

$$\int \left(1 + \frac{a^2 \sin^2 x}{2} \right) dx \rightarrow \int 1 dx + \frac{a^2}{2} \int \sin^2 x dx$$

$$\int 1 dx = x$$

$$\int \sin^2 x dx \rightarrow \int (1 - \cos(2x))/2 dx \rightarrow 1/2 \cdot \int (1 - \cos(2x)) dx$$

$$u = 2x, du = 2 dx, du/2 = dx$$

$$\frac{1}{2} \cdot \int 1 dx = \frac{1}{2} \cdot \int \cos(u) \cdot \frac{du}{2} \rightarrow \frac{x}{2} + \frac{1}{4} \sin(2x)$$

Integral junta

$$x + \frac{a^2}{2} \left(\frac{x}{2} + \frac{\sin(2x)}{4} \right)$$

Intervalo $0, \pi/2$

$$\frac{\pi}{2} + \left(\frac{a^2}{2} \cdot \frac{\pi}{2} + \frac{\sin(\pi)}{4} \right) - \left(0 + \frac{a^2}{2} \cdot \left(0 + \frac{\sin(0)}{4} \right) \right)$$

$$\frac{\pi}{2} + \left(\frac{a^2}{2} \cdot \left(\frac{\pi}{2} + 0 \right) \right) = 0$$

$$\frac{\pi}{2} + (\pi \cdot a^2) / 8 \rightarrow \frac{\pi}{2} \left(1 + \frac{a^2}{4} \right)$$

$$T = 2\sqrt{\frac{L}{g}} \cdot \frac{\pi}{2} \left(1 + \frac{a^2}{4} \right) \rightarrow 2\pi \sqrt{\frac{L}{g}} \cdot \left(1 + \frac{a^2}{4} \right)$$

$$T^2 = 4\pi^2 \cdot \frac{L}{G} \cdot \left(1 + \frac{a^2}{2} + \frac{a^4}{16}\right) \rightarrow g = \frac{4\pi^2 L}{T^2} \cdot \left(1 + \frac{a^2}{2} + \frac{a^4}{16}\right)$$

$$g = L\omega^2 \left(1 + \frac{a^2}{2} + \frac{a^4}{16}\right) \quad g = L \cdot \left(\omega^2 \cdot \left(1 + \frac{a^2}{2} + \frac{a^4}{16}\right)\right)$$

24) Seja $A(x,t) = A_0 \cos(Kx - \omega t)$ a função que descreve a posição x no instante t de uma onda com amplitude máxima A_0 , número de onda K e frequência angular ω . Mostre que a média temporal de $|A(x,t)|^2$ dentro de um período de oscilação

$T = 2\pi/\omega$ é igual a $1/2 A_0^2$, isto é, $\frac{1}{T} \int_0^T |A(x,t)|^2 dt = 1/2 A_0^2$.

$$A(x,t) = A_0 \cos(Kx - \omega t) \rightarrow \frac{1}{T} \int_0^T |A(x,t)|^2 dt = \frac{A_0^2}{2}$$

$$A(x,t)^2 = A_0^2 \cos^2(Kx - \omega t), \quad \phi = Kx \rightarrow A(x,t)^2 = A_0^2 \cos^2(\phi - \omega t)$$

$$\cos^2(\phi - \omega t) = \frac{1 + \cos(2\phi - 2\omega t)}{2}$$

$$\frac{1}{T} \int_0^T A_0^2 \cos^2(\phi - \omega t) dt = A_0^2 \cdot \frac{1}{T} \int_0^T \frac{1 + \cos(2\phi - 2\omega t)}{2} dt$$

$$\frac{A_0^2}{2} \cdot \frac{1}{T} \int_0^T (1 + \cos(2\phi - 2\omega t)) dt \rightarrow \frac{1}{T} \int_0^T A(x,t)^2 dt = \frac{A_0^2}{2}$$

$$\text{A média temporal é } \frac{1}{T} \int_0^T |A(x,t)|^2 dt = \frac{A_0^2}{2}$$

5) Calcule $\int \sin x \cdot \cos x \, dx$ por 4 métodos

a) $u = \cos x$

$$\int \sin x \cdot \cos x \, dx, \quad u = \cos x, \quad du = -\sin x \, dx, \quad -du = \sin x \, dx$$

$$\int u \cdot (-du) \rightarrow -\frac{u^2}{2} \rightarrow -\frac{\cos^2 x}{2} + C$$

b) $u = \sin x$

$$\int \sin x \cdot \cos x \, dx, \quad u = \sin x, \quad du = \cos x \, dx$$

$$\int u \cdot du \rightarrow \frac{u^2}{2} + C \rightarrow \frac{\sin^2 x}{2} + C$$

c) $\sin 2x = 2 \sin x \cdot \cos x$

$$\int \sin x \cdot \cos x \, dx = \frac{\sin 2x}{2} \rightarrow \frac{1}{2} \int \sin(2x) \, dx$$

1) $\int \sin(2x) \, dx, \quad u = 2x, \quad du = 2 \, dx, \quad du/2 = dx$

1) $\int \sin u \cdot \frac{1}{2} \cdot du \rightarrow \frac{1}{2} \int \sin u \, du$

1) $-\cos u + C \rightarrow -\cos x + C$

d) Integração por partes

$$\int \sin x \cdot \cos x \, dx, \quad u = \sin x, \quad du = \cos x \, dx, \quad dv = \cos x \, dx, \quad v = \sin x$$

Aplicando fórmula

$$\int u \cdot dv = \sin x \cdot \sin x - \int \sin x \cdot \cos x \, dx$$

$$\int \sin x \cdot \cos x \, dx = +\sin^2 x - \int \sin x \cdot \cos x \, dx$$

$$I = +\sin^2 x - I \rightarrow 2I = +\sin^2 x$$

$$I = \frac{+\sin^2 x}{2} + C$$

⑥ Calcule as integrais de produtos de funções trigonométricas com acas diferentes.

a) $\int \sin 8x \cdot \cos 5x \, dx$

$\sin 8x \cdot \cos 5x \, dx \rightarrow \frac{1}{2} [\sin(8x+5x) + \sin(8x-5x)]$

$\frac{1}{2} [\sin(13x) + \sin(3x)]$

$\frac{1}{2} (\sin(13x)) + \frac{1}{2} \sin(3x)$

$\frac{1}{2} \int \sin(13x) dx + \frac{1}{2} \int \sin(3x) dx$

$\sin(13x) dx = u = 13x, du = 13 dx, du/13 = dx$

$\int \sin u \cdot \frac{1}{13} du \rightarrow \frac{1}{13} \int \sin u \, du \rightarrow -\frac{1}{13} \cdot \cos(13x) dx$

$\int \sin(3x) dx = u = 3x, du = 3 dx, du/3 = dx$

$\int \sin(3x) dx = \int \sin u \cdot \frac{1}{3} dx \rightarrow -\frac{1}{3} \cdot \cos(3x) dx$

Voltando

$(-\frac{1}{2} \cdot \frac{1}{13} \cdot \cos(13x)) + (-\frac{1}{2} \cdot \frac{1}{3} \cdot \cos(3x))$

$-\frac{1}{26} \cdot \cos(13x) - \frac{1}{6} \cdot \cos(3x) + C$

b) $\int \sin 2x \cdot \cos 3x \, dx$

$\sin 2x \cdot \cos 3x \, dx \rightarrow \frac{1}{2} [\sin(2x+3x) + \sin(2x-3x)] dx$

$\frac{1}{2} [\sin(5x) + \sin(-x)] dx \rightarrow \frac{1}{2} \int \sin(5x) dx + \frac{1}{2} \int \sin(-x) dx$

$\sin(5x) dx = u = 5x, du = 5 dx, du/5 = dx$

$\int \sin(u) \cdot \frac{1}{5} du \rightarrow \frac{1}{5} \cdot -\cos(u) \rightarrow -\frac{1}{5} \cdot \cos(u)$

$\int \sin(-x) dx = u = -x, du = -dx, -du$

$\int \sin(u) \cdot -du \rightarrow +\cos(u)$

Voltando

$(-\frac{1}{2} \cdot \frac{1}{5} \cdot \cos(5x)) + (\frac{1}{2} \cdot \cos(x))$

$(-\frac{1}{10} \cdot \cos(5x)) + (\frac{1}{2} \cdot \cos(x))$

$-\frac{1}{10} \cdot \cos(5x) + \frac{1}{2} \cdot \cos(x) + C$

$$c) \int \sin 5x \cdot \sin x \, dx$$

$$1/2 \cdot [\cos(5x-x) - \cos(5x+x)]$$

$$\int 1/2 (\cos(4x)) - 1/2 (\cos(6x)) \rightarrow 1/2 \int \cos(4x) dx - 1/2 \int \cos(6x) dx$$

$$\int \cos(4x) dx \rightarrow u=4x, du=4dx, du/4=dx$$

$$\int \cos(u) \cdot 1/4 \cdot dx \rightarrow 1/4 \cdot -\sin(u) \rightarrow -1/4 \cdot \sin(4x)$$

$$\int \cos(6x) dx \rightarrow u=6x, du=6dx, du/6=dx$$

$$\int \cos(u) \cdot 1/6 \cdot du \rightarrow 1/6 \cdot -\sin(u) \rightarrow -1/6 \cdot \sin(6x)$$

Juntado

$$(1/2 \cdot -1/4 \cdot \sin(4x)) - (1/2 \cdot -1/6 \cdot \sin(6x))$$

$$-\frac{1}{8} \cdot \sin(4x) + \frac{1}{12} \cdot \sin(6x) + C$$

$$d) \int_{-\pi}^{\pi} \sin 2x \cdot \sin 3x \, dx$$

$$1/2 \cdot [\cos(2x-3x) - \cos(2x+3x)] dx$$

$$\int 1/2 \cdot \cos(-x) - 1/2 \cdot \cos(5x) \rightarrow 1/2 \int \cos(-x) dx - 1/2 \int \cos(5x) dx$$

$$\int \cos(-x) dx \rightarrow u=-x, du=-dx$$

$$-\int \cos u dx \rightarrow +\sin(x)$$

$$\int \cos 5x dx \rightarrow u=5x, du/5=dx$$

$$\int \cos u \cdot 1/5 \cdot du \rightarrow 1/5 \cdot \sin(5x)$$

Juntado

$$(1/2 \cdot \sin(x)) - (1/2 \cdot 1/5 \cdot \sin(5x))$$

$$+\frac{\sin(x)}{2} - \frac{\sin(5x)}{10} + C$$

Intervalo de $-\pi, \pi$.

$$\left(\frac{\sin(\pi)}{2} - \frac{\sin(5\pi)}{10} \right) - \left(\frac{\sin(-\pi)}{2} - \frac{\sin(-5\pi)}{10} \right)$$

$$0 - 0$$

$$e) \int \cos \pi x \cdot \cos 3\pi x \cdot dx$$

$$\int 1/2 \cdot [\cos(\pi + 3\pi) + \cos(\pi - 3\pi)] dx$$

$$\int 1/2 \cdot \cos(4\pi) + \int 1/2 \cdot \cos(-2\pi) \rightarrow 1/2 \cdot \int \cos(4\pi) + 1/2 \cdot \int \cos(-2\pi)$$

$$\int \cos(4\pi) dx \rightarrow u = 4\pi, du = 4 dx$$

$$\int \cos u \cdot \frac{1}{4} du \rightarrow 1/4 \cdot \sin(4\pi)$$

$$\int \cos(-2\pi) dx \rightarrow u = -2\pi, du = -2 dx$$

$$\int \cos(u) \cdot \frac{1}{-2} du \rightarrow -\frac{1}{2} \cdot \sin(-2\pi)$$

Juntado

$$1/2 \cdot (1/4 \cdot \sin(4\pi)) + 1/2 \cdot (-1/2 \cdot \sin(-2\pi))$$

$$1/10 \cdot \sin(4\pi) + 1/6 \cdot \sin(-2\pi) + C$$

$$\frac{\sin(5\pi x)}{10\pi} + \frac{\sin(3\pi x)}{6\pi} + C$$

$$f) \int_{-\pi/2}^{\pi/2} \cos x \cdot \cos 7x dx$$

$$\int 1/2 \cdot [\cos(x + 7x) + \cos(x - 7x)] dx$$

$$1/2 \int \cos(8x) + 1/2 \int \cos(-6x) dx$$

$$\int \cos(8x) \rightarrow u = 8x, du = 8 dx, du/8$$

$$\int \cos u \cdot 1/8 du \rightarrow 1/8 \cdot \sin(8x)$$

$$\int \cos(-6x) \rightarrow u = -6x, -du/6 = dx$$

$$\int \cos(u) \cdot \frac{1}{-6} du \rightarrow -\frac{1}{6} \cdot \sin(-6x)$$

$$1/2 \cdot (1/8 \cdot \sin(8x)) + 1/2 \cdot (-1/6 \cdot \sin(-6x))$$

$$1/16 \cdot \sin(8x) + 1/12 \cdot \sin(6x) + C$$

Intervalo de $-\pi/2, \pi/2$

$$\frac{\sin(4\pi)}{16} + \frac{\sin(3\pi)}{12} - \frac{\sin(-2\pi)}{16} + \frac{\sin(-3\pi)}{12}$$

$$0$$

$$0$$

$$0$$

$$0$$

$$0$$

7) Calcule a integral usando a substituição trigonométrica indicada.
Esboce o triângulo retângulo associado, indicando o ângulo θ e os comprimentos dos lados.

a) $\int x^3 \sqrt{9-x^2} dx$ ($x = 3 \sin \theta$)

$$\int x^3 \sqrt{9-x^2} \rightarrow x = 3 \sin \theta, dx = 3 \cos \theta d\theta$$

$$\sqrt{9-x^2} = 3 \cos \theta$$

Aplicando substituição

$$\int 27 \sin^3 \theta \cdot 3 \cos \theta \cdot 3 \cos \theta d\theta$$

$$\int 243 \sin^3 \theta \cos^2 \theta d\theta \rightarrow 243 \int \sin^2 \theta \cos^2 \theta \cos \theta d\theta$$

$$243 \int \sin^2 \theta \cos^2 \theta d\theta \rightarrow 243 \int \sin \theta (\sin^2 \theta) \cos^2 \theta d\theta$$

$$243 \int \sin \theta (1 - \cos^2 \theta) \cos^2 \theta d\theta$$

$$243 \int (\cos^2 \theta - \cos^4 \theta) \sin \theta d\theta \quad u = \cos \theta, du = -\sin \theta d\theta, -du = \sin \theta d\theta$$

$$-243 \int (u^2 - u^4) du \rightarrow -243 \int u^2 + 243 \int u^4 du$$

$$-\frac{243}{3} u^3 + \frac{243}{5} u^5 \rightarrow u = \cos \theta = \left(-\frac{\cos^3 \theta}{3} + \frac{\cos^5 \theta}{5} \right)$$

~~$\int x^3 \sqrt{9-x^2} dx$~~ $\left\{ \begin{array}{l} x = 3 \sin \theta \rightarrow \sin \theta = \frac{x}{3} \\ \cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \frac{x^2}{9}} \rightarrow \frac{\sqrt{9-x^2}}{3} \end{array} \right.$

Aplicando tudo

$$243 \left(-\frac{\cos^3 \theta}{3} + \frac{\cos^5 \theta}{5} \right) \rightarrow 243 \left(-\frac{1}{3} \cdot \left(\frac{\sqrt{9-x^2}}{3} \right)^3 + \frac{1}{5} \cdot \left(\frac{\sqrt{9-x^2}}{3} \right)^5 \right) + C$$

$$b) \int \frac{1}{x^2 \sqrt{x^2 - 9}} dx \quad x = 3 \sec \theta$$

$$\int \frac{1}{x^2 \sqrt{x^2 - 9}} dx \rightarrow u = 3 \sec \theta, du = 3 \sec \theta \cdot \tan \theta \cdot d\theta$$

$$\sqrt{u^2 - 9} = 3 \tan \theta$$

$$\int \frac{1}{9 \sec^2 \theta \cdot 3 \tan \theta} \cdot 3 \sec \theta \cdot \tan \theta \cdot d\theta \rightarrow \frac{1}{9} \int \frac{\sec \theta \cdot \tan \theta}{\sec^2 \theta \cdot \tan \theta} d\theta$$

$$\frac{1}{9} \int \frac{1}{\sec \theta} d\theta \rightarrow \frac{1}{9} \int \cos \theta d\theta \rightarrow \frac{1}{9} \cdot \sin \theta$$

Voltar para x .

$$x = 3 \sec \theta, \sec \theta = x/3$$

$$\cos \theta = \frac{1}{\sec \theta} \rightarrow \cos \theta = \frac{1}{x/3} \rightarrow \cos \theta = \frac{3}{x} \rightarrow \cos \theta = \frac{3}{x}$$

$$\sin^2 \theta = (1 - \cos^2 \theta) \rightarrow \sin^2 \theta = (1 - \frac{9}{x^2}) \quad \sin \theta = \frac{\sqrt{x^2 - 9}}{x}$$

$$\frac{\sin \theta}{\sec \theta} = \frac{\sqrt{x^2 - 9}}{x} \rightarrow \sin \theta = \frac{\sqrt{x^2 - 9}}{x}$$

Aplicando substituição: $\frac{1}{9} \cdot \frac{\sqrt{x^2 - 9}}{x} \rightarrow \frac{\sqrt{x^2 - 9}}{9x} + C$

$$c) \int \frac{x^3}{\sqrt{x^2 + 9}} dx$$

$$\int \frac{x^3}{\sqrt{x^2 + 9}} dx \rightarrow \int \frac{x^3}{\sqrt{x^2 + 3^2}} dx \rightarrow u = 3 \tan \theta, du = 3 \cdot \sec^2 \theta d\theta$$

$$\sqrt{u^2 - 9} = 3 \sec \theta$$

$$\int \frac{3 \tan^3 \theta}{3 \sec \theta} \cdot 3 \sec^2 \theta d\theta \rightarrow 27 \int \tan^3 \theta \cdot \sec \theta d\theta \rightarrow 27 \int \tan \cdot (\sec^2 - 1) \cdot \sec \theta d\theta$$

$$\int (\sec^3 \theta - \sec \theta) \cdot \tan \theta d\theta \rightarrow \int \sec^3 \theta \cdot \tan \theta - 3 \int \sec \theta \cdot \tan \theta$$

1º Integral

$$3 \sec^3 \theta \cdot \tan \theta \quad u = \sec, du = \sec \cdot \tan dx$$

$$\int u^3 \cdot \sec \cdot \tan \theta d\theta \rightarrow \int u^3 du \rightarrow \frac{u^4}{4} \rightarrow \frac{\sec^4 \theta}{4}$$

2º Integral

$$\int \sec \theta \cdot \tan \theta = \sec \theta$$

$$27 \left(\frac{\sec^3 \theta}{3} - \sec \theta \right)$$

Mudando para X .

$$X = 3 \tan \theta, \quad \tan \theta = X/3.$$

$$\sec^2 X = 1 + \tan^2 X \rightarrow \sec^2 X = \left(1 + \frac{X^2}{9}\right) \rightarrow \frac{X^2 + 9}{9}$$

$$\frac{\sqrt{\sec^2}}{\sqrt{3^3}} = \frac{\sqrt{X^2 + 9}}{\sqrt{27}} \rightarrow \sec \frac{\sqrt{X^2 + 9}}{3}$$

Voltando

$$27 \left(\frac{1}{3} \cdot \left(\frac{\sqrt{X^2 + 9}}{3} \right)^3 - \frac{\sqrt{X^2 + 9}}{3} \right)$$

8) Calcule a integral usando substituição trigonométrica.

$$a) \int_0^{3/2} \frac{dx}{\sqrt{9-x^2}} \quad X = 3 \sin \theta, \quad dx = 3 \cos \theta d\theta \quad \sqrt{9-x^2} = 3 \cos \theta$$

$$\int_0^{3/2} \frac{dx}{\sqrt{9-x^2}} \rightarrow \int_0^{3/2} \frac{3 \cos \theta d\theta}{3 \cos \theta} \quad \int_0^{3/2} d\theta \rightarrow \theta + C$$

$$X = 3 \sin \theta \rightarrow \frac{1}{\sin} = \frac{X}{3} \rightarrow \arcsin\left(\frac{X}{3}\right)$$

Intervalos

$$\arcsin\left(\frac{3}{3}\right) - \arcsin(0) \rightarrow \arcsin\left(\frac{1}{1}\right) - 0 = \frac{\pi}{6}$$

Portanto o resultado é $\frac{\pi}{6}$.

$$b) \int_0^3 \frac{x}{\sqrt{36-x^2}} dx \quad u = 6 \cdot \sin \theta, du = 6 \cdot \cos \theta dx, \sqrt{6^2 - x^2} = 6 \cdot \cos \theta$$

$$\int_0^3 \frac{6 \cdot \sin \theta \cdot 6 \cdot \cos \theta}{6 \cdot \cos \theta} d\theta \rightarrow 6 \int_0^{\pi/2} \sin \theta d\theta$$

$$6 \cdot \cos \theta + C$$

Voltando para x

$$x = 6 \sin \theta, \sin \theta = x/6$$

$$\cos^2 \theta = (1 - \sin^2 \theta) \rightarrow \cos^2 \theta = (1 - x^2/36) \rightarrow \cos \theta = \sqrt{36-x^2}/6$$

$$\cos \theta = \sqrt{36-x^2}/6$$

$$6 \cdot \sqrt{36-x^2}/6 + C \rightarrow \sqrt{36-x^2} + C$$

Intervalo de 0 a 3

$$\sqrt{36-3^2} - \sqrt{36-0} \rightarrow \sqrt{27} - 6 \rightarrow (3\sqrt{3} - 6)$$

$$c) \int_0^1 x^3 \sqrt{1-x^2} dx \quad u = \sin \theta, du = \cos \theta d\theta, \sqrt{1-x^2} = \cos \theta$$

$$\int_0^1 (\sin \theta)^3 \cdot \cos \theta \cdot \cos \theta d\theta$$

$$\int_0^1 (\sin \theta)^3 \cdot (\cos \theta)^2 d\theta \rightarrow \int_0^1 (\sin \theta) \cdot \sin^2 \theta \cdot (\cos \theta)^2 d\theta$$

$$\int_0^1 (1 - \cos^2 \theta) \cdot \cos^2 \theta \cdot \sin \theta d\theta \rightarrow \int_0^1 (\cos^2 \theta - \cos^4 \theta) \cdot \sin \theta d\theta, u = \cos \theta; du = -\sin \theta d\theta$$

$$\int_0^1 (u^2 - u^4) \cdot (-du) \rightarrow -\int_0^1 u^2 du - \int_0^1 u^4 du$$

$$\left(-\frac{u^3}{3} + \frac{u^5}{5}\right) + C \rightarrow \left(-\frac{\cos^3 \theta}{3} + \frac{\cos^5 \theta}{5}\right) + C$$

Voltando PARA x

$$x = \sin \theta, \sin \theta = x$$

$$\cos^2 \theta = (1 - \sin^2 \theta) \rightarrow \cos^2 \theta = (1 - x^2) \rightarrow \cos \theta = \sqrt{1-x^2}$$

Voltando

$$\left(-\frac{\sqrt{1-x^2}^3}{3} + \frac{\sqrt{1-x^2}^5}{5}\right)$$

Aplicando Intervalo 0 a 1

$$0 + \frac{1}{3} - \frac{1}{5} \rightarrow \frac{5}{15} - \frac{3}{15} = \frac{2}{15}$$

$$d) \int_{\sqrt{2}}^2 \left(\frac{1}{t^3 \sqrt{t^2 - 1}} \right) dt \quad u = \sec \theta, du = \sec \theta \tan \theta d\theta, \sqrt{t^2 - 1} = \tan \theta$$

$$\int_{\sqrt{2}}^2 \frac{1}{(\sec \theta)^3 \cdot \tan \theta} \cdot \sec \theta \tan \theta d\theta = \int \frac{1}{\sec^2 \theta} d\theta$$

$$\int_{\sqrt{2}}^2 \cos^2 \theta d\theta \rightarrow \int_{-\sqrt{2}}^{\sqrt{2}} \frac{1 + \cos(2x)}{2} \cdot \frac{1}{2} d\theta + \frac{1}{2} \int_{\sqrt{2}}^2 d\theta + \frac{1}{2} \int_{-\sqrt{2}}^{\sqrt{2}} \cos(2x) d\theta$$

Primeira Integral

$$\frac{1}{2} \int_{-\sqrt{2}}^{\sqrt{2}} d\theta \rightarrow \frac{1}{2} \theta \dots$$

Segunda Integral

$$\frac{1}{2} \int_{-\sqrt{2}}^{\sqrt{2}} \cos(2x) d\theta \rightarrow u = 2x, du = 2 dx \rightarrow du/2 = dx$$

$$\frac{1}{2} \int_{-\sqrt{2}}^{\sqrt{2}} \cos u \cdot \frac{1}{2} \rightarrow \frac{1}{4} \int_{-\sqrt{2}}^{\sqrt{2}} \cos u du \rightarrow \frac{1}{4} \cdot \sin u \rightarrow \frac{1}{4} \cdot \sin \theta$$

Voltando

$$\dots \frac{1}{2} \theta + \frac{1}{4} \sin 2\theta$$

Voltando para x .

$$x = \sec \theta \rightarrow x = \frac{1}{\cos \theta} \rightarrow \cos \theta = 1/x \quad \left| \begin{array}{l} \sin^2 \theta = (1 - \cos^2 \theta) \\ \sin \theta = \frac{\sqrt{x^2 - 1}}{x} \end{array} \right.$$

Aplicando em $\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta$

$$\left(\frac{\sec^{-1} x}{2} + \left(\frac{1}{4} \cdot 2 \cdot \frac{\sqrt{x^2 - 1}}{x} \cdot \frac{1}{x} \right) \right)$$

$$\frac{\sec^{-1} x}{2} + \frac{1}{2} \cdot \frac{\sqrt{x^2 - 1}}{x^2} \rightarrow \frac{\sec^{-1} x}{2} + \frac{\sqrt{x^2 - 1}}{2x^2} + C$$

Intervalo de $2a\sqrt{2}$

$$\frac{\sqrt{2^2 - 1}}{2 \cdot 4} - \frac{\sqrt{2 - 1}}{2 \cdot 2} \rightarrow \frac{\sqrt{3}}{8} - \frac{1}{4}$$

e) ~~Resolva~~ $\int \frac{dx}{x^2 \sqrt{4+x^2}}$ $u = 2 \operatorname{tg} \theta$, $du = 2 \sec^2 \theta d\theta$, $2 \sec \theta$

$$\int \frac{2 \sec \theta d\theta}{(2 \operatorname{tg} \theta)^2 \cdot 2 \sec \theta} \rightarrow \frac{1}{24} \int \frac{\sec \theta d\theta}{\operatorname{tg} \theta^2} \rightarrow \frac{1}{24} \int \frac{\cos \theta d\theta}{\sin^3 \theta}$$

$u = \sin \theta$, $du = \cos \theta d\theta$

$$\frac{1}{24} \int \frac{\cos \theta d\theta}{u^3} \rightarrow \frac{1}{24} \int \frac{du}{u^3} \rightarrow \frac{1}{24} \int u^{-3} du \rightarrow \frac{1}{24} \cdot \frac{u^{-2}}{-2}$$

$$\frac{1}{24} \cdot -u^{-2} \rightarrow \frac{1}{24} \cdot -\frac{1}{u} \rightarrow \frac{1}{24} \cdot -\frac{1}{\sin \theta} + C$$

Voltando para x

$x = 2 \operatorname{tg} \theta \rightarrow \operatorname{tg} \theta = x/2$

cateto oposto = x / adjacente = 2 / Hip = $\sqrt{x^2+4}$

$\sin \theta = \frac{x}{\sqrt{x^2+4}}$

Doncundo $\frac{1}{24} \cdot -\frac{1}{\frac{x}{\sqrt{x^2+4}}} \rightarrow \frac{1}{24} \cdot -\frac{\sqrt{x^2+4}}{x} + C \rightarrow \frac{-\sqrt{x^2+4}}{24x} + C$

f) $\int \frac{x^2}{\sqrt{x^2-4}} dx$ $u = 2 \sec \theta$, $du = 2 \sec \theta \operatorname{tg} \theta d\theta$, $2 \operatorname{tg} \theta$

$\int \frac{4 \sec^2 \theta \cdot 2 \sec \theta \operatorname{tg} \theta d\theta}{2 \sec \theta} \rightarrow \int 4 \sec^3 \theta d\theta$

Integral $\sec^3 \theta$

$\left(\frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| \right) \cdot 4$

$2 \sec \theta \tan \theta + 2 \ln |\sec \theta + \tan \theta|$

$\sec \theta = x/2$

$\tan^2 \theta = (\sec^2 \theta - 1) \rightarrow \left(\frac{x^2}{4} - 1 \right) \rightarrow \sqrt{\frac{x^2-4}{4}} = \tan \theta$

Voltando

$\left(2 \cdot \frac{x}{2} \cdot \frac{\sqrt{x^2-4}}{2} \right) + 2 \ln \left| \frac{x}{2} + \frac{\sqrt{x^2-4}}{2} \right|$

$\frac{x \cdot \sqrt{x^2-4}}{2} + 2 \ln \left| \frac{x}{2} + \frac{\sqrt{x^2-4}}{2} \right|$

$$g) \int \frac{x^2}{\sqrt{9-25x^2}} dx$$

$$\int \frac{x^2}{\sqrt{9-(5x)^2}} \rightarrow \int \frac{x^2}{3\sqrt{1-(\frac{5x}{3})^2}} \rightarrow \frac{1}{3} \int \frac{x^2}{\sqrt{1-(\frac{5x}{3})^2}} du$$

$$5x = 3 \sin \theta \rightarrow x = \frac{3}{5} \sin \theta \quad || \quad dx = \frac{3}{5} \cos \theta d\theta \quad || \quad \sqrt{1-(\frac{5x}{3})^2} \rightarrow \sqrt{1-\sin^2 \theta} \rightarrow \cos \theta$$

Aplicando Substituição

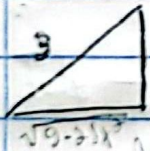
$$\frac{1}{3} \int \frac{9 \frac{1}{25} \sin^2 \theta \cdot \frac{3}{5} \cos \theta d\theta}{\cos \theta} \rightarrow \frac{27}{375} \int \sin^2 \theta d\theta \rightarrow \frac{9}{125} \int \sin^2 \theta d\theta$$

$$\int \sin^2 \theta d\theta \rightarrow \int \frac{1 - \cos(2\theta)}{2} d\theta \rightarrow \frac{1}{2} \theta - \frac{1}{2} \int \cos u \cdot \frac{du}{2} \rightarrow \frac{1}{2} \theta - \frac{1}{4} \sin u$$

$$\frac{1}{2} \int 1 dx - \frac{1}{2} \int \cos(2x) dx \rightarrow \frac{1}{2} \theta - \frac{1}{2} \int \cos u \cdot \frac{du}{2} \rightarrow \frac{1}{2} \theta - \frac{1}{4} \sin u$$

$$\frac{1}{2} \theta - \frac{1}{4} \sin(2\theta) + C$$

$$\frac{9}{125} \cdot \left(\frac{1}{2} \theta - \frac{1}{4} \sin 2\theta \right) \rightarrow \frac{9}{250} \cdot \theta + \frac{9}{500} \sin 2\theta$$



$$\sin \theta = \frac{5x}{3} \quad || \quad \cos \theta = \frac{\sqrt{9-25x^2}}{3} \quad || \quad \tan \theta = \frac{5x}{\sqrt{9-25x^2}}$$

Aplicando

$$\frac{9}{250} \arcsin\left(\frac{5x}{3}\right) - \frac{9}{500} \cdot \frac{5x \cdot 2 \cdot \sqrt{9-25x^2}}{3}$$

$$\frac{9}{250} \arcsin\left(\frac{5x}{3}\right) - \frac{3}{50} \cdot \frac{5x \cdot \sqrt{9-25x^2}}{3}$$

$$\frac{9}{250} \arcsin\left(\frac{5x}{3}\right) - \frac{x \cdot \sqrt{9-25x^2}}{50} + C$$

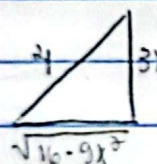
$$b) \int dx / \sqrt{16-9x^2} + \int dx / 4\sqrt{1-(3x/4)^2} \rightarrow || \int dx / \sqrt{1-(3x/4)^2} ||$$

$$3x/4 = \sin \theta \rightarrow x = 4/3 \sin \theta \quad || \quad dx = 4/3 \cos \theta \quad || \quad \sqrt{16-(3x)^2} \rightarrow 4 \cos \theta$$

Aplicando Substituição

$$\int \frac{4/3 \cos \theta d\theta}{4 \cos \theta} \quad S = \frac{4/3 d\theta}{4} = \frac{1}{3} \int d\theta \rightarrow \frac{1}{3} \theta + \frac{\theta/3}{1/3} + C$$

Voltando para X.



$$\sin = \frac{3x}{4} \quad \cos = \frac{\sqrt{16-9x^2}}{4} \quad \tan = \frac{3x}{\sqrt{16-9x^2}}$$

$$\int dx / \sqrt{16-9x^2} = \frac{1}{3} \arcsin(3x/4)$$

$$1) \int x / \sqrt{x^2+4x-5} \quad \int x / \sqrt{(x+2)^2-9} = 1$$

$$x+2 = 3 \sec \theta \rightarrow x = (3 \sec \theta - 2) \quad || \quad dx = 3 \sec \theta \cdot \tan \theta d\theta$$

$$\sqrt{(3 \sec \theta)^2 - 9} = 3 \tan \theta$$

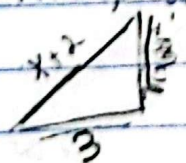
Aplicando Substituição

$$\int \frac{(3 \sec \theta - 2) \cdot 3 \sec \theta \tan \theta d\theta}{3 \tan \theta} \rightarrow 3 \int (\sec \theta - 2) \cdot \sec \theta d\theta \rightarrow 3 \int \sec^2 \theta - 2 \sec \theta d\theta$$

$$3 \int \sec^2 \theta d\theta - 2 \int \sec \theta d\theta \quad || \text{Ambas integrais "Diretas"}$$

$$3 \cdot \tan \theta - 2 \ln | \sec \theta + \tan \theta | + C$$

Triângulo



$$\sec = \frac{\sqrt{(x+2)^2-9}}{x+2} \quad \cos = \frac{3}{x+2} \quad \tan = \frac{\sqrt{(x+2)^2-9}}{3}$$

$$3 \cdot \frac{\sqrt{(x+2)^2-9}}{3} - 2 \ln | x+2/3 + \sqrt{(x+2)^2-9}/3 | + C$$

$$\sqrt{(x+2)^2-9} - 2 \ln | x+2 + \sqrt{(x+2)^2-9} | + C$$

$$J) \int \sqrt{x^2+2x} \, dx = \int \sqrt{(x^2+2x+1)-1} \, dx = \int \sqrt{(x+1)^2-1} \, dx$$

$$(x+1) = \sec \theta \rightarrow x = \sec \theta - 1 \parallel dx = \sec \theta \cdot \tan \theta \, d\theta$$

$$\sqrt{\sec^2 \theta - 1} = \tan \theta$$

Aplicando Substituição:

$$\int \tan \theta \cdot \sec \theta \cdot \tan \theta \, d\theta \rightarrow \int \sec \theta \cdot \tan^2 \theta \, d\theta$$

$$\int \sec \theta \cdot (1 - \sec^2 \theta) \, d\theta \rightarrow \int \sec \theta - \sec^3 \theta \, d\theta$$

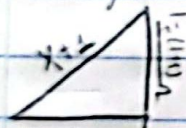
$$\int \sec \theta \, d\theta - \int \sec^3 \theta \, d\theta$$

$$1^\circ \rightarrow \int \sec \theta \, d\theta = \ln |\sec \theta + \tan \theta|$$

$$2^\circ \rightarrow \int \sec^3 \theta \, d\theta \rightarrow \frac{1}{2} \sec(x) \cdot \tan(x) + \frac{1}{2} \ln |\sec x + \tan x| + C$$



Triângulo



Verfando

$$\sec \theta = \frac{\sqrt{(x+1)^2-1}}{(x+1)} \quad \cos = \frac{1}{x+1} \quad \tan = \frac{\sqrt{(x+1)^2-1}}{1}$$

$$\frac{1}{2} \cdot (x+1) \cdot (\sqrt{(x+1)^2-1}) + \frac{1}{2} \cdot \ln |x+1 + \sqrt{(x+1)^2-1}| +$$

$$\text{Total} \rightarrow \ln |x+1 + \sqrt{(x+1)^2-1}| - \frac{(x+1)}{2} \cdot (\sqrt{(x+1)^2-1}) + \ln |x+1 + \sqrt{(x+1)^2-1}| + C$$

$$K) \int dx / \sqrt{x^2-6x+3} \quad \int dx / \sqrt{(x-3)^2-4} \quad \int dx / \sqrt{(x-3)^2+4}$$

$$(x-3) = 2 \tan \theta = x = 2 \tan \theta + 3 \parallel dx = 2 \sec^2 \theta \, d\theta$$

$$\sqrt{4 \tan^2 \theta + 4} \rightarrow 2 \sec \theta$$

Aplicando Substituição:

$$\int \frac{2 \sec^2 \theta \, d\theta}{2 \sec \theta} \rightarrow \int \sec \theta \, d\theta \rightarrow \ln |\sec \theta + \tan \theta| + C$$

Triângulo



Verfando Para X

$$\sec \theta = \frac{x-3}{2} \quad \cos = \frac{2}{x-3} \quad \tan \theta = \frac{\sqrt{(x-3)^2-4}}{2}$$

$$\ln \left| \frac{\sqrt{(x-3)^2-4}}{2} + \frac{x-3}{2} \right| + C$$

$$1) \int \sqrt{8-2x-x^2} dx \rightarrow \int \sqrt{-x^2+2x+9} \rightarrow \int \sqrt{-(x+1)^2+9}$$

$$(x+1)=3\cos\theta \rightarrow x=3\cos\theta-1 \quad || \quad dx = -3\sin\theta d\theta \quad || \quad \sqrt{-(x+1)^2+9} \rightarrow \sqrt{-(3\cos\theta-1)^2+9} = 3\sin\theta$$

Aplicando SUBSTITUIÇÃO

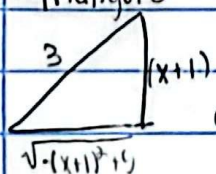
$$\int 3\cos\theta \cdot 3\sin\theta d\theta \rightarrow 9 \int \cos\theta \sin\theta d\theta \rightarrow 9 \int \frac{1}{2} (1+\cos 2\theta) d\theta$$

$$9/2 \int (1+\cos 2\theta) d\theta \quad u=2\theta, du=2d\theta, du/2=d\theta$$

$$9/2 \int (1 d\theta) + \int (\cos u) d\theta \rightarrow 9/2 \theta + 9/2 \int \cos u \cdot \frac{1}{2} du$$

$$9/2 \theta + 9/4 \sin 2\theta \rightarrow$$

Triângulo



$$\sin = \frac{(x+1)}{3} \quad \cos = \frac{\sqrt{-(x+1)^2+9}}{3}$$

Voltando para X

$$\left\{ \begin{array}{l} 9/2 \cdot \sin\theta \cdot ((x+1)/3) + 9/4 \cdot 2 \cdot (x+1) \cdot \sqrt{-(x+1)^2+9} / 2 \\ \frac{9}{2} \cdot \sin\theta \cdot \frac{(x+1)}{3} + \frac{9}{2} \cdot (x+1) \cdot \frac{\sqrt{-(x+1)^2+9}}{2} \end{array} \right.$$

- 9) Uma circunferência de raio R centrada na origem do plano cartesiano tem equação $x^2+y^2=R^2$. Mostre que a área da região delimitada pela circunferência é πR^2 .

$$y = \sqrt{R^2-x^2} \rightarrow A = \int_{-R}^R \sqrt{R^2-x^2} dx$$

$$x = R \sin(\theta), dx = R \cos(\theta) d\theta$$

Tracar limites de Integração.

$$x = -R \rightarrow -\pi/2 \quad || \quad x = R \rightarrow \pi/2$$

$$A = 2 \int_{-R}^R \sqrt{R^2-x^2} dx = 2 \int_{-\pi/2}^{\pi/2} \sqrt{R^2-R^2 \sin^2(\theta)} \cdot R \cos\theta d\theta$$

$$A = 2 \int_{-\pi/2}^{\pi/2} R \cdot (\cos\theta) \cdot R \cos\theta d\theta = 2R^2 \int_{-\pi/2}^{\pi/2} \cos^2\theta d\theta$$

$$A = 2R^2 \int_{-\pi/2}^{\pi/2} \frac{1+\cos(2\theta)}{2} d\theta = R^2 \int_{-\pi/2}^{\pi/2} (1+\cos(2\theta)) d\theta$$

$$A = R^2 \cdot [\theta]_{-\pi/2}^{\pi/2} = R^2 \cdot \pi$$

10- Resolva a integral $\int \frac{\tan[\ln(x + \sqrt{1+x^2})]}{\sqrt{1+x^2}} dx$

$$\ln(x + \sqrt{1+x^2}) = \operatorname{arsh}(x)$$

$$\int \frac{\tan(\operatorname{arsh}(x))}{\sqrt{1+x^2}} dx \quad u = \operatorname{arsh}(x) \quad x = \sinh(u) \quad dx = \cosh(u) du$$

$$\frac{\int \tan(u)}{\sqrt{1+\sinh^2(u)}} \rightarrow \frac{\int \tan(u)}{\cosh(u)} \cdot \cosh(u) du \rightarrow \int \tan(u) du$$

$$\int \tan(u) du = -\ln|\cos(u)| + C \rightarrow -\ln|\cos(\ln(x + \sqrt{1+x^2}))| + C$$